

Introduction to Retrieval-Augmented Generation

Abstract

Retrieval-Augmented Generation (RAG) is a technique that lets a Large Language Model answer questions using a corpus of documents it was never trained on. The technique combines a retrieval step over a knowledge base with a generation step performed by the language model. RAG was first introduced by Lewis et al. in 2020 as a way to ground language model outputs in verifiable sources.

1. The Problem RAG Solves

Large Language Models have two well-known failure modes when asked about topics outside their training distribution. First, they hallucinate — producing confident text that has no basis in fact. Second, they are limited by their knowledge cutoff date and cannot answer questions about events, products, or documents that did not exist at training time. RAG addresses both problems by injecting relevant source material into the model's context at query time.

2. The Five Stages of a RAG Pipeline

A standard RAG pipeline consists of five sequential stages: ingest, chunk, embed, retrieve, and generate. The first three stages run at index time, typically once when the corpus is updated. The last two stages run at query time, once per user question. The separation between index-time and query-time work is what allows RAG systems to scale to millions of documents while still answering questions in milliseconds.

3. Chunking and Why It Matters

A chunk is a small contiguous piece of a source document. Chunks exist for two reasons. First, language model context windows are finite — even a 200,000-token window cannot hold a million-document corpus. Second, retrieval is more precise on small chunks than on full documents, because an embedding of a 50-page document averages out the meaning of every page. A typical chunk size is 500 to 1500 characters with 10-20 percent overlap between neighbours, but optimal sizes are domain-specific.

4. Grounding the Generation Step

After retrieval returns the top-K most-relevant chunks, the generation step composes a prompt of the form: SYSTEM = answer using only the context below. CONTEXT = retrieved chunks. QUESTION = the user query. This prompt structure, combined with explicit instructions to refuse to answer when the context is insufficient, is what reduces hallucination. Grounding is not automatic — it must be requested in the prompt.

5. Failure Modes

RAG systems still fail in three predictable ways. Out-of-corpus questions return no relevant chunks and require the model to refuse honestly. Ambiguous questions retrieve the wrong chunks. And retrieval misses occur when the embedding model fails to map a paraphrased query to the correct chunk. Sessions 21 through 28 of this fellowship address each of these failure modes in turn.

References

Lewis, P. et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS 2020.